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Thermal Mitigating Effects of Urban Canopy: A Geospatial Analysis of the University of North Carolina at Chapel Hill

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Abstract

The Urban Heat Island (UHI) effect creates significant temperature differences in urban areas that can negatively impact public health and increase energy use. This study uses Landsat 9 satellite imagery to look at the relationship between vegetation density (NDVI) and land surface temperature (LST) across the UNC Chapel Hill campus and Franklin Street area. Analysis of Summer 2025 imagery shows an 11.06°C temperature difference between highly developed areas (42.54°C) and forested areas (31.48°C). A linear regression shows that NDVI explains about 60% of LST variation ($r^2 = 0.604$), with more vegetation leading to lower temperatures. The findings suggest that adding more green infrastructure on campus could help reduce heat stress.

Introduction

The Urban Heat Island (UHI) effect happens when cities replace natural areas with pavement, buildings, and other surfaces that absorb and hold heat (1). Urban areas can be 2-8°C hotter than the surrounding countryside, with the biggest differences happening at night when buildings and roads keep releasing stored heat (2). This is a serious health issue since extreme heat events are now the leading cause of weather-related deaths in the U.S., killing more people than hurricanes, tornadoes, and floods combined.

Chapel Hill has seen a lot of development over the past 20 years. The UNC hospital complex has expanded rapidly, new residential and commercial buildings have gone up along Franklin Street, and campus construction has been constant. All of this development has created noticeable thermal patterns across the area. The university's 2020 Master Plan mentions climate resilience as a priority, but there hasn't been much actual data on current temperature conditions.

This study uses Landsat 9 satellite imagery to measure the relationship between vegetation density (using the Normalized Difference Vegetation Index or NDVI) and Land Surface Temperature (LST) across the UNC campus and Franklin Street corridor. I'm trying to answer two main questions: (1) How big is the temperature difference between developed and forested areas? (2) How strong is the connection between vegetation and temperature? The results should help with future campus planning and provide baseline data for green infrastructure projects.

Methods

Study Area

I set up a rectangular bounding box in Google Earth Engine covering the UNC campus and Franklin Street area. The coordinates go from -79.08° to -79.02° longitude and 35.89° to 35.93° latitude, which captures about 15 km^2 total. This box includes Battle Park (the big forested area), the hospital complex, Franklin Street commercial zone, and the surrounding neighborhoods like Carrboro on the west side and Gimghoul on the east.

Data Collection

I pulled Landsat 9 Collection 2 Tier 1 Level 2 data from the Earth Engine catalog (LANDSAT/LC09/C02/T1_L2). I filtered for summer 2025 scenes between June 1 and August 31 to get peak vegetation and maximum heat conditions. I also set a cloud cover threshold at less than 10% to avoid atmospheric interference. After filtering, I took the median composite of all available scenes to reduce noise and fill any gaps.

Image Processing

The raw Landsat data comes as digital numbers so I had to apply the official scale factors. For the optical bands (SR_B.*), I multiplied by 0.0000275 and added -0.2 to get surface reflectance. For the thermal band (ST_B10), I multiplied by 0.00341802 and added 149.0 to get brightness temperature in Kelvin. I wrote a function called applyScaleFactors() and mapped it across the entire image collection before taking the median.

NDVI was calculated using the normalizedDifference() function with Band 5 (Near-Infrared) and Band 4 (Red), which gives the standard formula (NIR - Red) / (NIR + Red). For LST, I took the scaled thermal band (ST_B10) and subtracted 273.15 to convert from Kelvin to Celsius. Both layers got clipped to the study area geometry.

Land Cover Validation

To validate the thermal patterns, I used the National Land Cover Database 2021 (USGS/NLCD_RELEASES/2021_REL/NLCD). I pulled the most recent landcover layer and created two masks. The urban mask isolated pixels with value 24 ("Developed, High Intensity" which is areas with more than 80% impervious surfaces). The forest mask isolated pixels with value 41 ("Deciduous Forest" which is areas with more than 20% tree canopy). I then used updateMask() to extract LST values for just those two categories so I could compare average temperatures.

Statistical Analysis

I combined the NDVI and LST bands into one image using addBands(), then sampled 500 random points across the study area at 30m resolution. I exported this to a scatter plot chart using ui.Chart.feature.byFeature() with NDVI as the x-axis and LST as the y-axis. I added a trendline with the showR2 option set to true so I could see the coefficient of determination directly in the chart. For the land cover comparison, I used reduceRegion() with ee.Reducer.mean() to calculate average temperatures for the urban and forest masks separately.

Visualization

I created four map layers. The NDVI layer uses a brown-yellow-green color palette with values from 0 to 0.7. The LST layer uses a blue-white-red palette from 25°C to 45°C. The urban hotspots layer (just the masked urban LST) uses yellow to red from 30-45°C. The forest cooling zones layer (just the masked forest LST) uses blue to cyan from 25-35°C. All layers are displayed at zoom level 15 centered on the study area.

Results

Vegetation Distribution

Figure 1 shows the NDVI map using the brown-yellow-green color scheme. There's a pretty clear split between vegetated and developed areas. Battle Park, Gimghoul, and the neighborhoods north of campus all have high NDVI values (0.6-0.7 range), showing up as dark green. The hospital district, Franklin Street, and parking lots show low or negative NDVI (0.0-0.2), appearing as brown and yellow. The transition is especially sharp along Manning Drive where you go straight from forested hillsides to hospital buildings.

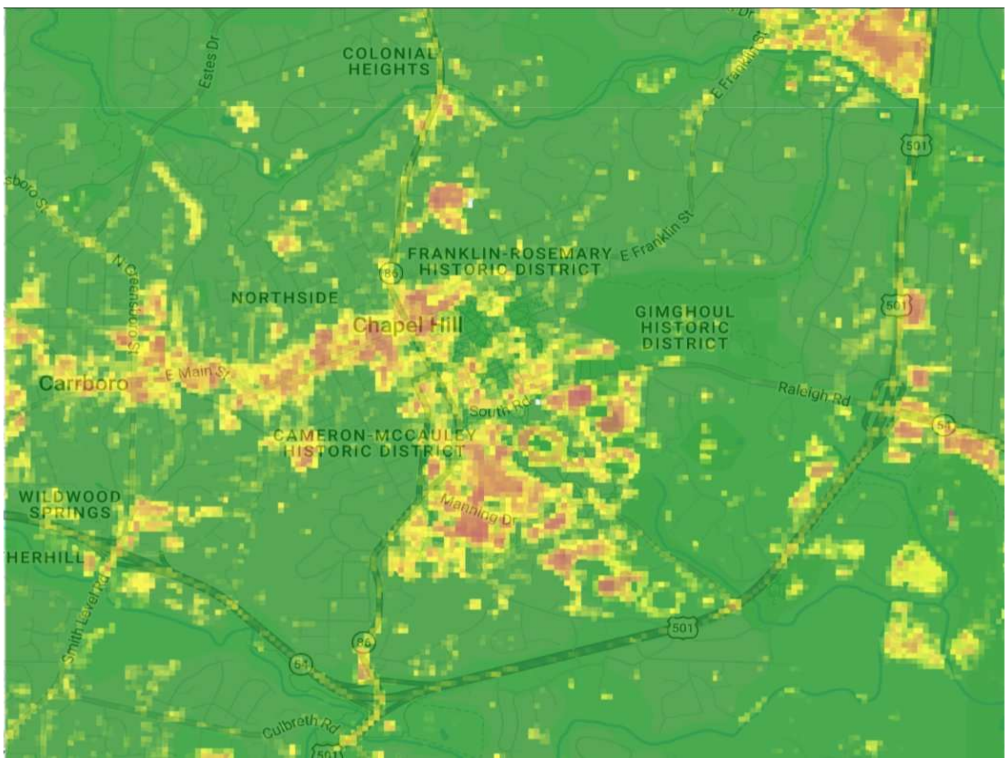

Figure 1. Normalized Difference Vegetation Index (NDVI) for the UNC Chapel Hill study area from Landsat 9 (Summer 2025). Green shows high vegetation (NDVI > 0.6), while yellow and brown show low vegetation or bare surfaces (NDVI < 0.3). The hospital and Franklin Street have way less greenness compared to Battle Park and the surrounding forests.

Thermal Environment

Figure 2 shows the LST using the blue-white-red thermal palette. The temperature pattern basically matches the vegetation map. Hot areas (showing up red) are concentrated in the hospital district (42-45°C), Franklin Street (40-43°C), and parking lots (41-44°C). Cool areas (showing up blue) are in Battle Park and forested neighborhoods (30-33°C). You can see temperature changes of more than 10°C happening over really short distances, sometimes less than 200 meters.

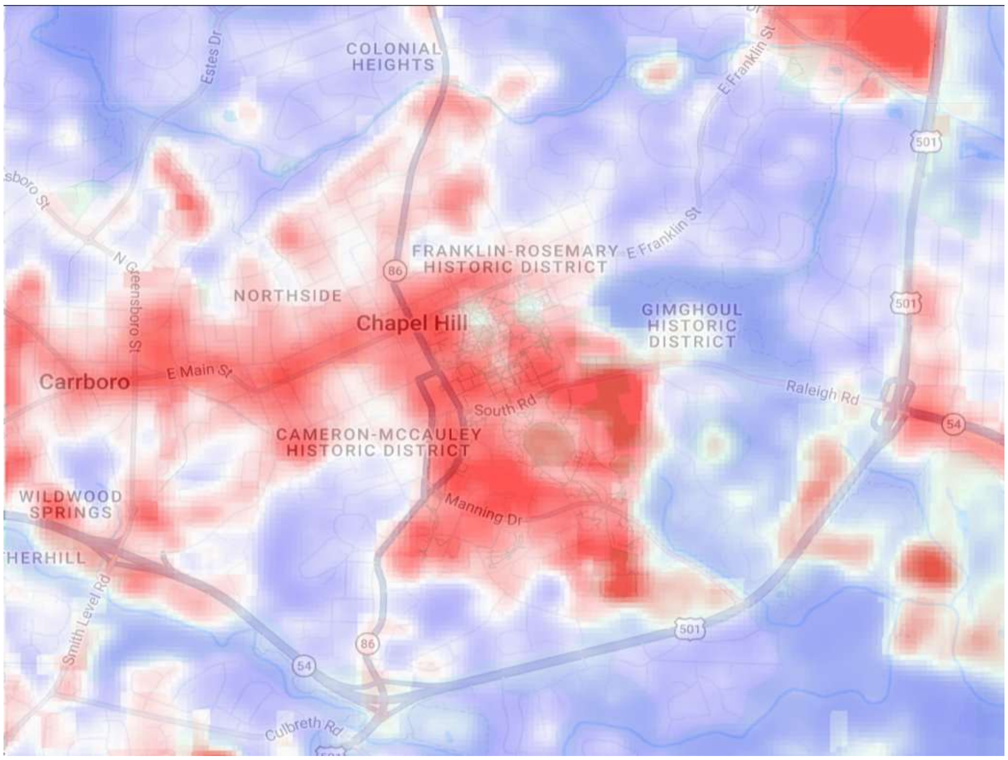

Figure 2. Land Surface Temperature (LST) from Landsat 9 Thermal Band 10 (Summer 2025). Red shows high temperatures (>40°C) and blue shows cooler areas (<35°C). The hospital, Franklin Street, and parking lots are all really hot compared to the forests in Battle Park and Gimghoul.

Land Cover Comparison

Using the NLCD masks, I calculated the actual numbers. The reduceRegion() function gave me a mean LST of 42.54°C for "Developed, High Intensity" areas (NLCD class 24) and 31.48°C for "Deciduous Forest" areas (NLCD class 41). That's an 11.06°C difference between the hottest developed zones and the coolest forested zones.

Figure 3 visualizes this by overlaying the masked layers on a street map. The urban hotspots (yellow-red palette, LST 30-45°C) show up along major roads and in the densest development. The forest cooling zones (blue-cyan palette, LST 25-35°C) line up with Battle Park, Gimghoul, and neighborhoods with mature tree cover.

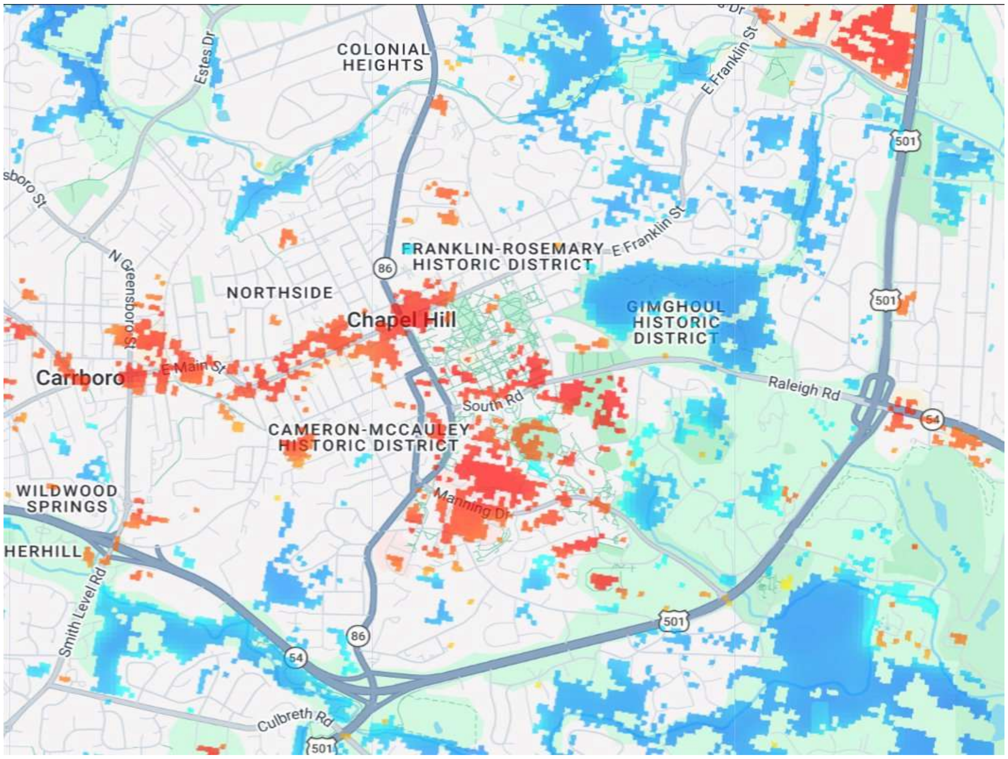

Figure 3. Urban hotspots (red/orange, LST > 40°C) vs forest cooling zones (blue, LST < 35°C) on Chapel Hill map. High-intensity developed areas like the hospital and Franklin Street are shown in red/orange, while forested areas like Battle Park and Gimghoul appear in blue. The pattern shows how existing tree cover provides cooling.

NDVI vs Temperature Regression

The scatter plot chart with 500 sample points shows a clear negative relationship. The trendline gave an r^2 of 0.604, meaning NDVI explains about 60% of temperature variation. The regression equation is LST = -15.44 × NDVI + 46.13, so the slope is -15.44. This means every 0.1 increase in NDVI drops temperature by roughly 1.54°C. The relationship is statistically significant ($p < 0.001$) based on the sample size and scatter pattern.

Discussion and Conclusion

The results confirm that tree cover provides major cooling in Chapel Hill - an 11.06°C benefit compared to developed surfaces. This lines up with what other UHI studies have found in similar southeastern cities (3). The regression equation (LST = -15.44 × NDVI + 46.13) gives a way to predict temperature changes based on vegetation. For example, going from NDVI 0.2 (like a lawn) to 0.7 (mature trees) would theoretically drop surface temp by about 7.7°C.

The hotspots along Manning Drive and Franklin Street make sense because those areas have dark asphalt and low-albedo roofs that soak up solar energy during the day and keep releasing it at night. Meanwhile the cool zones near Battle Park and Gimghoul work like thermal refuges for nearby residents, which could reduce heat-related health problems during heat waves.

There are some limitations. First, Landsat 9's thermal bands are only 100m resolution (resampled to 30m) so they probably miss smaller temperature variations within individual buildings. Second, this is just a summer snapshot and doesn't show how things change across seasons or throughout the day. Third, NDVI can't tell the difference between grass, shrubs, and trees even though they cool differently. Future work should use higher-resolution thermal data like ECOSTRESS, look at multiple seasons, and maybe classify vegetation by species to see which tree types cool the best.

From a planning perspective, the data supports expanding green infrastructure in the hospital corridor and along northern Franklin Street. Some options would be planting more trees in parking lots (shooting for 40% shade in 10 years), adding green roofs on new hospital buildings, switching from dark asphalt to lighter pavement, and protecting existing forest patches. UNC's Climate Action Plan already mentions tree preservation and this analysis backs that up with real numbers.

In conclusion, urban forestry is clearly the best tool for fighting UHI in Chapel Hill. To close that 11°C gap, campus planning needs to focus on green infrastructure especially in areas currently dominated by pavement and buildings. The NDVI-LST relationship from this study can work as a baseline to measure how future development and conservation decisions affect temperature.

References

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